

Deployment Guide

AccessGaurd System

1. Deployment Objectives

The main goal of this deployment is to create a fully automated structure which:

- Supplies the accounts of users immediately when the onboarding process begins
- Assigns users with predefined responsibilities and roles
- Enforces the security policies of the organization in the system
- When the users account is terminated, the deactivation process starts immediately
- Maintain an accurate and full audit log of all the supply activities

The process of deployment must make sure that each of the components are operating with one another in a secure manner within the main IT environment of the organization.

2. Pre-Deployment Requirements

Before the deployment process, the team must align the necessary foundational components to ensure a smooth deployment. The HR system needs to be properly configured in a way that can manage the user Lifecycle data while assigning roles, hiring, and creating the termination status. This structure will take the place of a primary trigger for the offboarding and onboarding workflow.

A centralized Access and identity management (IAM) structure needs to be in place to be able to look user the accounts of users and enforce proper authorization and authentication policies. This structure must support the controls that are role-based and manage secure credentials.

It is the responsibility of the organization to have proper service management software's to fast track the offboarding and onboarding activities through the ticketing workflow. In addition, a device management system shall be available to enforce compliance policies and manage the control endpoint devices.

Finally, a monitoring system and centralized logging need to be properly configured to analyze and capture the structure for audit and security purposes.

Required Tools & Technologies

Tool	Version	Purpose	Install Method
Docker	24.x	Container runtime for all services	apt-get install docker.io
Kubernetes (k8s)	1.28+	Container orchestration	kubeadm or managed k8s
Helm	3.x	Kubernetes package manager	snap install helm --classic
Keycloak	22.x	Auth and identity server	Helm chart: codecentric/keycloak
PostgreSQL	16.x	Primary relational database	Helm chart: bitnami/postgresql
Redis	7.x	Token Store & Session cache	Helm chart: bitnami/redis
HashiCorp Vault	1.15	Secrets management	Helm chart: hashicorp/vault
ELK Stack	8.x	SIEM & Log aggregation	Helm chart: elastic/elasticsearch
Wazuh	4.x	Threat detection	Docker Compose (wazuh/wazuh-docker)
Open Policy Agent	0.60+	RBAC policy engine	OPA Kubernetes admission webhook

Architecture Overview of Deployment

This structure uses a connected architecture in which the HR system will start the process of the automation layer. This layer will automatically communicate with the IAM structure to help interact with the device management tools, manage the account of the user, update the management system to track the progress, and manage the tools to control the endpoints.

Phase 1: Access and Identity management Setup

The initial deployment phase highlights the access management and identity management system configuration, serving as the main component of the user lifecycle. The structures of the user accounts must be defined such as the naming conventions stating the attributes of the employees properly like department, employee ID and job role. These are essential in role assignment and automated provisioning.

The role-based access control then needs to be added in by listing the roles that correspond with the job functions inside of the organization. Each of the roles need to have been given permissions which help to determine the amount of access the user has in the data and system. This makes sure that all users are gaining only the necessary amount of access to perform their normal duties.

The authentication policies must also be configured to use multi-factor authentication and strong password requirements. These policies are imperative for protecting the account of users from unauthorized access and protecting their accounts. Lastly, the logging needs to be enabled inside of the IAM system to capture the events that are all identity-related, such as the login attempts, account creation and permission changes.

Phase 2: HR System Integration

When the IAM system is properly configured, the next phase of integrating the HR system begins within the automation layer.

The HR system needs to be enabled to properly generate events when a new user is added, terminated, or updated. These events serve as the trigger points for the offboarding and onboarding workflow.

The data fields within the HR system need to be mapped with the corresponding fields in the IAM system. This can include the department, job title, employee name and status. Accurate mapping of data allows for proper creation of user accounts and assigned cases.

The integration between the automation layer and the HR system must be tested to ensure that each item runs successfully and receives and transmits data properly. These test cases should

always have the termination of an existing employee and a creation of a new employee on record.

Phase 3: Automation Workflow Deployment

The automation layer is then deployed to start the process of executing the offboarding and offboarding workflows as well as the HR events.

For the Onboarding process, the system needs to create the user accounts, generate service tickets, assign roles, and supply system access. Each of these steps need to be executed within the proper sequence making sure that all the users are given the proper resources upon joining the organization.

For the offboarding process, the system needs to be configured to disable the accounts of users, revoke permissions, terminate the active sessions and start the device management actions. This system needs to allow for each permission to be removed when the process of termination starts to keep away from unauthorized usage.

Phase 4: Device Management Deployment

The device management system ends to be configured to properly enforce the security policies within all the endpoint devices.

Each of the device need to be enrolled in the management system prior to them being issued as employees. The enrolment ensures that each of these devices can be controlled by the organization. Each of these devices need to be configured with the security policies such as encryptions, system updates and antivirus protection. These policies help to secure the device and ensures that it stays compliant with the standards of the organization.

During the process of offboarding, the system shall be capable of removing the corporate data from the device and revoking the access the user had. This makes sure that all the companies' sensitive information is not retained with the user after they leave the organization.

Phase 5: Service Management Integration

The service management system needs to be configured to properly track the offboarding and onboarding activities throughout the generation of tickets. When the process starts a ticket needs to be created to document the supplying of the process. This ticket needs to

include the user's relevant information to track the completion of such steps within the workflow.

During the offboarding process, the ticket needs to be generated to document the deactivation of the account as well as the device management actions and the access removal. These tickets help to provide a record for all the IT activities as well as promote accountability. This system needs to be integrated with the layer of automation to allow for the updated automatically to take place within the workflow.

Phase 6: Monitoring setup and security logging

The last and final phase of this deployment process involves for the proper configuration of the monitoring and security logging system. All components of this structure shall be configured to properly send off the logs to a logging platform that is centralized. This can include the automation workflows, identity management systems and device management tools. Monitoring rules need to be created to detect any suspicious activity like unauthorized access attempts, failed login attempts and incomplete offboarding process.

Alerts need to be configured to notify the administrators of the IT team when any similar event occurs. This makes sure that all of the potential security events are identified early and addressed.

The retention of logs policies needs to be defined to make sure that all of the data is stored in a safe location for an appropriate amount of time to help with the investigations and support.

System validation and testing

After the deployment takes place, run the checks from the validation checklist prior to opening the system to new users. Each of these checks need to pass before going live.

Authentication validation – Auth Smoke Tests

Test 1: login with a test user – expect MFA presented challenges and JWT issued

Test 2: Login with an invalid password 5 times – Expect the account to be locked after the 5th attempt.

Test 3: Use an expired token – Expect to be given a 401 unauthorized with a redirect to re-authorization.

Test 4: Access the endpoints without proper tokens – expect to see a 401 with no sensitive detail of the error exposed.

Test 5: RBAC check – the Standard user should not admin or access any of the endpoints, expect a 403.

Integration Validation – Integration checks

Keycloak -> PostgreSQL the data of the user is consistent across the Keycloak when restarting

Services -> Audit logger the events of authentication appear within 30 seconds in the Elasticsearch.

Audit Logger -> Kibana the dashboard for the log shows any real-time events

Wazuh -> Pager duty the alert triggers a test confirming the incident of the Pager Duty is created.

Vault -> Services the service gains a secret without manual intervention during the startup.

Redis -> Keycloak, the sessions that are stored in redis help to verify the blacklist tokens efforts during logout.

Security validation – Security Checks

TLS: Confirm that each of the endpoints return a valid TLS certificate with none of the HTTP being available.

CSP Headers: Make sure that all the admin dashboards return a content-security policy header

Secrets: make sure that there are no secrets within the environment container logs or variables.

RBAC: make sure that the OPA denies the policy evaluations which are unauthorized.

Backup: Make sure that the trigger backups manually, making sure the file is checked summed and written.

Recovery: restore the process from backing to staging, confirming the integrity of the data.